

1. [In this question, $\mathbf{i}$ and $\mathbf{j}$ are horizontal perpendicular unit vectors.]

A particle A has mass 2 kg and a particle B has mass 3 kg. The particles are moving on a smooth horizontal plane when they collide.

Immediately before the collision, the velocity of A is $5\mathbf{j} \text{ m s}^{-1}$ and the velocity of B is $(3\mathbf{i} - \mathbf{j}) \text{ m s}^{-1}$

Immediately after the collision, the velocity of A is $(3\mathbf{i} + 2\mathbf{j}) \text{ m s}^{-1}$

- (a) Find the total kinetic energy of the two particles before the collision.

(3)

- (b) Find, in terms of $\mathbf{i}$ and $\mathbf{j}$, the impulse received by A in the collision.

(2)

Given that, in the collision, the impulse of A on B is equal and opposite to the impulse of B on A ,

- (c) find the velocity of B immediately after the collision.

(3)

2.

In this question you must show all stages of your working.

Solutions relying on calculator technology are not acceptable.

A particle P is moving in a straight line.

At time t seconds, the speed, $v \text{ m s}^{-1}$, of P is given by the continuous function

$$v = \begin{cases} \sqrt{2t+1} & 0 \leq t \leq k \\ \frac{3}{4}t & t > k \end{cases}$$

where k is a constant.

(a) Show that $k = 4$, explaining your method carefully.

(3)

(b) Find the acceleration of P when $t = 1.5$

(3)

At time $t = 0$, P passes through the point O

(c) Find the distance of P from O when $t = 8$

(7)

3.

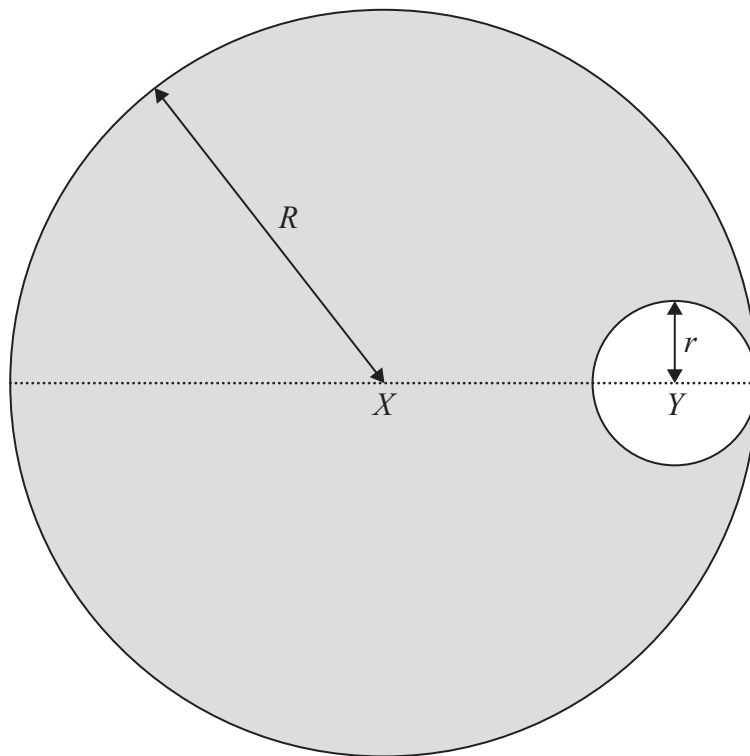


Figure 1

A uniform circular disc C has centre X and radius R .

A disc with centre Y and radius r , where $0 < r < R$ and $XY = R - r$, is removed from C to form the template shown shaded in Figure 1.

The centre of mass of the template is a distance kr from X .

(a) Show that $r = \frac{k}{1-k} R$ (4)

(b) Hence find the range of possible values of k . (2)

The point P is on the outer edge of the template and PX is perpendicular to XY .

The template is freely suspended from P and hangs in equilibrium.

Given that $k = \frac{4}{9}$

(c) find the angle that XY makes with the vertical. (3)

The mass of the template is M .

(d) Find, in terms of M , the mass of the lightest particle that could be attached to the template so that it would hang in equilibrium from P with XY horizontal. (3)

4. A rough plane is inclined to the horizontal at an angle α , where $\tan \alpha = \frac{3}{4}$

A particle P of mass m is held at rest at a point A on the plane.

The particle is then projected with speed u up a line of greatest slope of the plane and comes to instantaneous rest at the point B .

The coefficient of friction between the particle and the plane is $\frac{1}{7}$

- (a) Show that the magnitude of the frictional force acting on the particle, as it moves from A to B , is $\frac{4mg}{35}$ (2)

Given that $u = \sqrt{10ag}$, use the work–energy principle

- (b) to find AB in terms of a , (4)

- (c) to find, in terms of a and g , the speed of P when it returns to A . (4)
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5. A particle P of mass m and a particle Q of mass $2m$ are at rest on a smooth horizontal plane.

Particle P is projected with speed u along the plane towards Q and the particles collide. The coefficient of restitution between the particles is e .

As a result of the collision, the direction of motion of P is reversed.

- (a) Find, in terms of u and e , the speed of P after the collision.

(6)

After the collision, Q goes on to hit a vertical wall which is fixed at right angles to the direction of motion of Q . The coefficient of restitution between Q and the wall is $\frac{1}{3}$

Given that there is a second collision between P and Q

- (b) find the full range of possible values of e .

(5)

6.

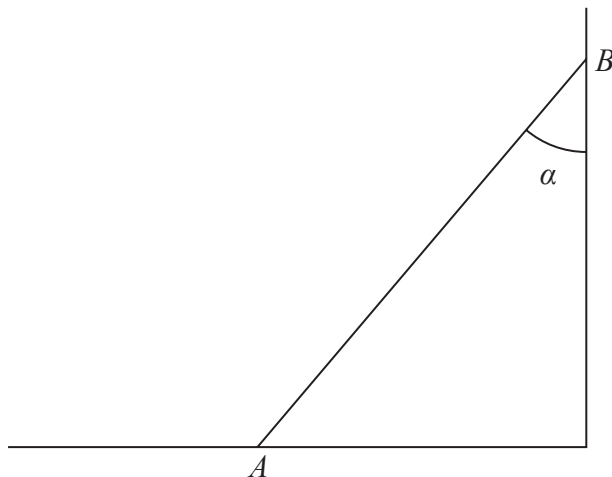


Figure 2

A uniform rod, AB , of mass m and length $2a$, rests in limiting equilibrium with its end A on rough horizontal ground and its end B against a smooth vertical wall.

The vertical plane containing the rod is at right angles to the wall.

The rod is inclined to the wall at an angle α , as shown in Figure 2.

The coefficient of friction between the rod and the ground is $\frac{1}{3}$

(a) Show that $\tan \alpha = \frac{2}{3}$

(6)

With the rod in the same position, a horizontal force of magnitude kmg is applied to the rod at A , towards the wall. The line of action of this force is at right angles to the wall.

The rod remains in equilibrium.

(b) Find the largest possible value of k .

(4)

7. [In this question, the unit vectors $\mathbf{i}$ and $\mathbf{j}$ are in a vertical plane, $\mathbf{i}$ being horizontal and $\mathbf{j}$ being vertically upwards.]

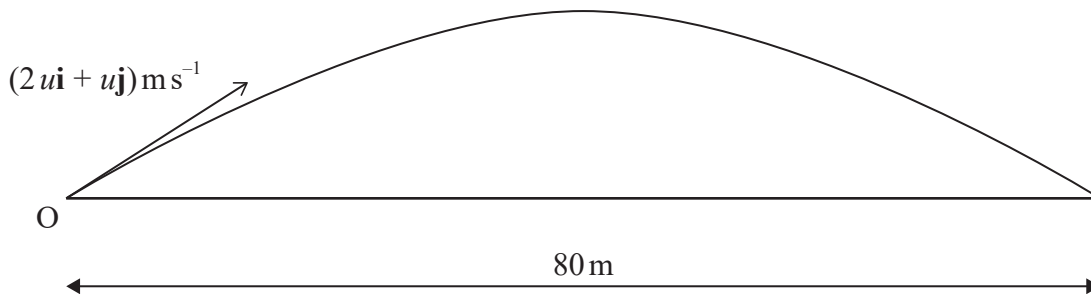


Figure 3

A golf ball is hit from a point O on horizontal ground and is modelled as a particle moving freely under gravity. The initial velocity of the ball is $(2u\mathbf{i} + u\mathbf{j})\text{ m s}^{-1}$. The ball first hits the horizontal ground at a point which is 80 m from O , as shown in Figure 3.

Use the model to

- (a) show that $u = 14$

(6)

- (b) find the total time, while the ball is in the air, for which the speed of the ball is greater than $7\sqrt{17}\text{ m s}^{-1}$

(5)